

ANALYSTLAB AFRICA
HEALTHCONNECT CLINIC EXPERIENCE LAB
WEEK 8: FINAL DATA ANALYTICS AND DECISION SUPPORT

Table 1

<u>Requirements</u>	<u>Response</u>
What is my final component?	HealthConnect Final Analytics Dashboard and Decision Support Report
What problem does it address?	Understanding patient appointment attendance and no-show patterns to support decisions aimed at reducing missed appointments and improving use of appointment slots.
What is its current status?	Final dashboard developed, tested and refined following Week 7 validation.
What did I improve during Week 7?	Dashboard presentation, visual clarity, KPI validation, filter functionality and interpretation of analytical findings.
Which other track(s) does it depend on?	Data science
Which track(s) depend on my output?	Data Science may use validated findings for model interpretation
What output will I provide?	Validated KPIs, dashboard visuals, attendance/no-show findings, business insights, recommendations and analytical limitations.
What output do I need from another track?	Enter the actual output exchanged during your cross-track activity.
What evidence demonstrates readiness?	Validated dashboard figures, tested slicers/filters, refined visuals and documented findings.
What limitations remain?	I didn't get a data science student to collaborate with.

Executive Summary

Week 8 represents the final Data Analytics stage of the HealthConnect project. The work continues directly from the Power BI dashboard developed and tested during Weeks 5,6 and 7. The objective is to convert the validated analytical work into a final decision support output that clearly communicates appointment attendance patterns, no-show patterns, business implications and evidence-based recommendations.

The final analysis uses 5,000 HealthConnect appointment records. The validated baseline shows a 48.46% no-show rate, 46.28% attendance rate, 5.26% cancellation rate and 72.68% reminder coverage. The analysis also identifies clear differences in no-show rates across booking lead time, previous no-show history, distance to the clinic and reminder status.

The strongest observed segmentation patterns are the higher no-show rate among appointments booked 31–60 days in advance (60.49%) and among patients with three or more previous no-shows (68.82%). Patients travelling 20+ km also show a higher no-show rate (57.76%). Appointments with reminders show a lower observed no-show rate than appointments without reminders (47.36% versus 51.39%). These are observational findings.

Final KPIs

The following KPIs were established and validated during the previous analytics work and are carried forward as the final Week 8 baseline.

- ✓ Total Appointments 5000 - these are the total appointments records analyzed.
- ✓ Total No Shows 2423 - Appointments recorded as No-Show.
- ✓ No Show Rate 46.48% - Share of appointments recorded as No-Show.
- ✓ Attendance Rate 46.28% - Share of appointments recorded as Attended.
- ✓ Cancellation Rate 5.26% - Share of appointments recorded as Cancelled.
- ✓ Reminder Coverage 72.68% - Share of appointments for which a reminder was recorded as sent.

Key Findings

1. Booking Lead Time

Lead Time Group	Appointments	No-Show Rate	Attendance Rate
0–7 days	640	27.81%	66.56%
8–14 days	602	33.55%	61.79%
15–30 days	1,333	43.21%	51.69%
31–60 days	2,425	60.49%	34.10%

Interpretation: The no-show rate increases across the lead-time groups, with the 31–60 day group showing the highest observed no-show rate and the lowest attendance rate among the listed groups.

2. Previous No Show History

Previous No-Shows	Appointments	No-Show Rate	Attendance Rate
0	2,921	43.51%	50.46%
1	1,548	53.49%	42.25%
2	438	59.36%	36.30%
3+	93	68.82%	29.03%

Interpretation: The observed no-show rate increases with previous no-show history. This supports using previous no-show history as a descriptive segmentation variable.

3. Distance to Clinic

Distance Group	Appointments	No-Show Rate	Attendance Rate
0–5 km	1,156	46.45%	48.96%
5.1–10 km	1,692	46.51%	47.75%
10.1–20 km	1,669	49.43%	45.12%
20+ km	393	57.76%	37.66%

Interpretation: The 20+ km group has the highest observed no-show rate and lowest attendance rate among the distance groups shown.

4. Reminder Status

Reminder Sent	Appointments	No-Show Rate	Attendance Rate
No	1,366	51.39%	42.68%
Yes	3,634	47.36%	47.63%

Interpretation: The reminder group has an observed no-show rate approximately 4.03 percentage points lower than the no-reminder group. This should be described as an association rather than evidence that reminders alone caused the difference.

Key Visualizations

The following visuals are relevant to the final Data Analytics output:

I. Clustered column charts

- ✓ Total Appointment by Appointment Outcome.
- ✓ Total Appointment by Previous No Show Group
- ✓ No-Show Rate by Reminder Sent.
- ✓ No-Show Rate by Lead Time Group.
- ✓ No-Show Rate by Previous No-Show Group.

- ✓ No-Show Rate by Distance Group.
- ✓ No-Show Rate by Appointment Type.
- ✓ No-Show Rate by Appointment Day.
- ✓ No-Show Rate by Appointment Outcome.
- ✓ No-Show Rate by Distance Group and Reminder Sent.

II. Line chart

- ✓ No-Show Rate by Previous No-Show Group

III. Donut chart

- ✓ No-Show Rate by Risk Segment.

IV. matrix

- ✓ No-Show Rate by Lead Time Group and Reminder Sent.

V. Slicers

- ✓ Age Group
- ✓ Gender
- ✓ Appointment types
- ✓ Appointment day
- ✓ Reminder sent

Business Insights

- I. High number of missed appointments
Almost half of the appointments in the dataset were missed, with a no-show rate of 48.46%. This shows that missed appointments are an important area for HealthConnect to monitor and address.
- II. Appointments booked further in advance had more no-shows
The analysis shows that appointments booked further ahead tended to have higher no-show rates. The 31–60 day group recorded the highest no-show rate at 60.49%, compared with 27.81% for appointments booked within 0–7 days.
- III. Previous no-shows were linked with higher observed no-show rates
Patients with a history of missing appointments had higher no-show rates in the dataset. The rate increased from 43.51% among patients with no previous no-shows to 68.82% among those with three or more previous no-shows.
- IV. Patients travelling longer distances showed higher no-show rates
The analysis found a higher no-show rate among patients travelling longer distances. Patients travelling 20 km or more had a 57.76% no-show rate, compared with 46.45% among those travelling 0–5 km.
- V. Reminder status showed a difference in no-show rates
Appointments where a reminder was recorded had a 47.36% no-show rate, compared with 51.39% when no reminder was recorded. This suggests that reminder status is an important factor to monitor when analyzing appointment attendance.

- VI. Reminder patterns were also visible across booking periods
The analysis showed that appointments with reminders generally had lower observed no-show rates across the different booking lead-time groups. However, this is an observed relationship and does not by itself prove that reminders caused the lower no-show rate.

Actionable Recommendations

- I. Focus on reducing the overall no-show rate
The clinic should regularly monitor missed appointments and use the dashboard to identify areas where no-shows are highest. This can help the clinic track changes and make better decisions.
- II. Pay more attention to appointments booked far in advance
Since appointments booked 31–60 days ahead had the highest no-show rate; the clinic could give these appointments additional attention through follow-ups or appointment confirmations closer to the scheduled date.
- III. Identify patients with a history of missed appointments
Patients who have previously missed appointments showed higher no-show rates. The clinic could use this information to identify patients who may need additional appointment reminders or follow-up.
- IV. Consider travel distance when planning appointment follow-ups
Patients travelling longer distances showed higher no-show rates. Distance can therefore be considered when reviewing appointment patterns and planning ways to improve attendance.

Analytical Limitations

- I. The HealthConnect dataset is a project dataset representing a fictional clinic scenario rather than a real clinic population.
- II. The analysis is observational and identifies associations; it does not establish causation.
- III. The available data cannot explain why an individual patient missed an appointment when that reason is not recorded.
- IV. The analysis is limited to the variables available in the approved HealthConnect appointment dataset.
- V. The intended Data Science cross track validation was not completed because an available Data Scientist was not identified.